MareHealth MSS Alert GUI v3EN — Quick Start Guide

MareHealth-dynamic MSS Alert System v3.1

Somatic + Psychological Dual-Dimension Monitoring | UVP-TOF-MS Exhaled Breath Metabolomics

1. Overview

MareHealth-dynamic MSS Alert System is a Windows GUI application for real-time metabolic stress monitoring of seafarers based on UVP-TOF-MS exhaled breath metabolomics.It combines two scoring branches:

- **MSS-physo**: somatic symptom risk (fidget, respiratory, dizziness, physical, digestive, insomnia), graded GREEN / YELLOW / BLUE / RED with staged logic;
- **MSS-psycho**: psychological state prediction (fatigue, emotion, stress) from metabolic delta (Δm/z) relative to the previous measurement.

No Python installation is required. The program runs as a single executable.

2. System Requirements

- Windows 10 / 11 (64-bit)
- Input data file: Excel, or CSV containing Subject ID, Time, the 20 m/z abundance columns, plus optional questionnaire columns (fatigue_total, emotion_total, stress_total) and somatic symptom columns
- ~200 MB free disk space (temporary files during unpacking on first run)

3. Quick Start

1. Double-click MareHealth_MSS_Alert_GUI_v3_EN.exe.
2. In the **Control Panel**:

- **Data File**: click *Browse* and select your data file (default: final_merged_data_Top20.xlsx).
- **Output Dir**: click *Browse* and choose where results will be saved.

3. Click **Start Batch Analysis**. Wait until the log shows Done!.
4. In the **Radar Chart Visualization** panel:

- Select a **Subject** and a **Date**.
- Click **Single-Timepoint Radar** for one-timepoint somatic + psycho radar charts, or **All-Timepoint Comparison** for the full longitudinal radar array.

5. Right-click the chart to **Save as PNG / PDF / SVG** (300 dpi, publication-ready).

4. Understanding the Output

- **Overall Risk (4-tier, staged)**:
- GREEN — normal; maintain routine monitoring.
- YELLOW — monitor closely; increase measurement frequency.
- BLUE — early warning (baseline under construction, 1st–2nd measurement); result for reference only, repeat within 24–48 h.
- RED — immediate attention required.
- **Radar charts**: red dashed ring = risk score 0.5 (HIGH), orange = 0.3 (ELEVATED). Psychological axes show N/A for a subject's first measurement (no previous spectrum).
- **Info panel**: MSS score, cluster phenotype, predicted vs. measured questionnaire scores, trend summary, and intervention recommendations.

Output files (in the selected Output Dir)

File	Contents
marehealth_dynamic_alert_results.csv	Flattened results for all measurements
marehealth_dynamic_alert_reports.txt	Full human-readable alert reports
marehealth_sample_alert_result.json	Machine-readable sample result

5. Troubleshooting

Problem	Solution
Nothing happens on double-click	Run from Command Prompt to see the error; add the exe to antivirus whitelist
"File not found" during batch analysis	Re-select the data file with *Browse*; avoid Chinese characters / spaces in the path
Blank psycho radar (N/A)	Expected for each subject's first timepoint; psycho scoring starts at the 2nd measurement
Chart does not appear	Confirm the batch analysis finished; click a subject/date again

Designed by Dr. Xuelu Ding, Dr. Shuai Liang, and colleagues — 2026